

Recall from last class:

- A real **vector space** is a non-empty set V on which two operations, scalar multiplication and addition, are defined subject to the following 10 axioms. We call elements in V vectors. Let $u, v, w \in V$ and $c, d \in \mathbb{R}$.
 1. $u + v \in V$.
 2. $u + v = v + u$.
 3. $(u + v) + w = u + (v + w)$.
 4. there exists a vector 0_V in V such that $u + 0_V = u$.
 5. there exists a vector $(-u)$ in V such that $u + (-u) = 0_V$.
 6. $cu \in V$.
 7. $c(u + v) = cu + cv$.
 8. $(c + d)u = cu + du$.
 9. $c(du) = (cd)u$.
 10. $1 \cdot u = u$.
- A **subspace** of a vector space V is a subset of V such that
 1. $0_V \in H$.
 2. H is closed under vector addition; i.e. if $u, v \in H$, then $u + v \in H$.
 3. H is closed under scalar multiplication; i.e. if $c \in \mathbb{R}$ and $u \in H$, then $cu \in H$.
- **Theorem 4.1** If $v_1, \dots, v_p \in V$, then $\text{Span}\{v_1, \dots, v_p\}$ is a subspace of V .
- We call $\text{Span}\{v_1, \dots, v_p\}$ the subspace generated by or spanned by $v_1, \dots, v_p$.

Definition 1. *The null space of an $m \times n$ matrix A is*

$$\text{Null}(A) = N(A) = \{x \in \mathbb{R}^n : A\vec{x} = \vec{0}\}.$$

Theorem: If A is an $m \times n$ matrix, then $N(A)$ is a subspace of $\mathbb{R}^n$.

1. Prove the Theorem.

2. Let $\vec{b}$ be a non-zero element in $\mathbb{R}^m$. Is $\{\vec{x} \in \mathbb{R}^n : A\vec{x} = \vec{b}\}$ a subspace of $\mathbb{R}^n$?

3. Let $A = \begin{bmatrix} 1 & -4 & 0 & 2 & 0 \\ 0 & 0 & 1 & -5 & 0 \\ 0 & 0 & 0 & 0 & 2 \end{bmatrix}$. Find $N(A)$.

Definition 2. The **column space** of an $m \times n$ matrix A is the span of the columns of A . We denote the column space of A by $\text{Col}(A)$.

Theorem: If A is an $m \times n$ matrix, then $\text{Col}(A)$ is a subspace of $\mathbb{R}^m$.

4. Let $A = \begin{bmatrix} 1 & -4 & 0 & 2 & 0 \\ 0 & 0 & 1 & -5 & 0 \\ 0 & 0 & 0 & 0 & 2 \end{bmatrix}$. Find $\text{Col}(A)$.

Definition 3. Let V and W be vector spaces, then $T : V \rightarrow W$ is a **linear transformation** if

(a) $T(u + v) = T(u) + T(v)$ for all $u, v \in V$ and

(b) $T(cu) = cT(u)$ for all $c \in \mathbb{R}$ and $u \in V$.

Definition 4. The **range or image** of T is defined by

$$\text{Range}(T) = \text{im}(T) = \{T(x) : x \in V\} = \{w \in W : \text{there exists } a \in V \text{ such that } T(a) = w\}.$$

The **kernel** of T is defined by

$$\ker(T) = \{x \in V : T(x) = 0_W\}.$$

5. Define $T : \mathbb{P}_n \rightarrow \mathbb{P}_{n-1}$ by $T(f) = \frac{df}{dt}$.

(a) Is T linear?

(b) Find the image of T .

(c) Find the kernel of T .

6. Define $T : \mathbb{P}_2 \rightarrow \mathbb{P}_3$ by $T(p) = tp(t) + p'(t)$.

(a) Find the range of T .

(b) Find the kernel of T .

(c) Is T onto? Is T one-to-one?

7. Let $T : \mathbb{P}_2 \rightarrow \mathbb{R}^2$ by $T(p) = \begin{bmatrix} p(0) \\ p(1) \end{bmatrix}$.

(a) Is T linear?

(b) Find $\ker(T)$ and $\text{Range}(T)$.

8. Define $T : \mathbb{P}_2 \rightarrow M_{2 \times 2}$ by $T(p) = \begin{bmatrix} p(1) - p(0) & 0 \\ 0 & p(0) \end{bmatrix}$. Find the kernel and image of T .

Null Spaces/Column Spaces – Answers / Solutions

1. *Proof.* Let A be an $m \times n$ matrix, then $N(A) \subseteq \mathbb{R}^n$. We check the three subspace conditions.

(a) Since $A \cdot 0 = 0$, $0 \in N(A)$.

(b) Let $x, y \in N(A)$, then $A(x + y) = Ax + Ay = 0 + 0 = 0$. Thus $(x + y) \in N(A)$.

(c) Let $x \in N(A)$ and $c \in \mathbb{R}$, then $A(cx) = c(Ax) = c(0) = 0$. Thus $cx \in N(A)$.

Thus $N(A)$ is a subspace of $\mathbb{R}^n$. □

2. No, the set is not a subspace. Since $A0 \neq b$, $0 \notin \{x : Ax = b\}$. Thus the first subspace condition fails.
3. Fortunately, A is nearly in reduced row echelon form. By definition $\vec{x} \in N(A)$ if and only if $A\vec{x} = \vec{0}$. Now $A\vec{x} = \vec{0}$ if and only if

$$\begin{aligned}x_1 - 4x_2 + 2x_4 &= 0 \\x_3 - 5x_4 &= 0 \\2x_5 &= 0.\end{aligned}$$

Thus $\vec{x} \in N(A)$ if and only if

$$\vec{x} = \begin{bmatrix} 4x_2 - 2x_4 \\ x_2 \\ 5x_4 \\ x_4 \\ 0 \end{bmatrix} = x_2 \begin{bmatrix} 4 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} + x_4 \begin{bmatrix} -2 \\ 0 \\ 5 \\ 1 \\ 0 \end{bmatrix}.$$

Thus $\vec{x} \in N(A)$ if and only if

$$\vec{x} \in \text{Span} \left\{ \begin{bmatrix} 4 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} -2 \\ 0 \\ 5 \\ 1 \\ 0 \end{bmatrix} \right\}.$$

Therefore,

$$N(A) = \text{Span} \left\{ \begin{bmatrix} 4 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} -2 \\ 0 \\ 5 \\ 1 \\ 0 \end{bmatrix} \right\}.$$

4. The column space of A is defined to be the span of the columns of A . Thus

$$\text{Col}(A) = \text{Span} \left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} -4 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 2 \\ -5 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 2 \end{bmatrix} \right\}.$$

Notice that some of the vectors in the span are redundant, so we can express the column space of A using only the first, third, and fifth vectors; namely,

$$\text{Col}(A) = \text{Span} \left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 2 \end{bmatrix} \right\}.$$

5. (a) We check the two linearity conditions. Let $c \in \mathbb{R}$ and $f, g \in \mathbb{P}_n$. Note

$$T(f + g) = \frac{d}{dt}(f + g) = \frac{df}{dt} + \frac{dg}{dt}$$

from the sum rule for differentiation. Thus $T(f + g) = T(f) + T(g)$. Now we check T respects scalar multiplication:

$$T(cf) = \frac{d}{dt}(cf) = c \frac{df}{dt}$$

from the constant multiple differentiation rule. Thus $T(cf) = cT(f)$

Since the two conditions hold, T is linear.

- (b) Suppose $f \in \mathbb{P}_{n-1}$, then $f(t) = a_0 + a_1t + \cdots + a_{n-1}t^{n-1}$. Antidifferentiating we get $T(a_0t + \frac{a_1}{2}t^2 + \cdots + \frac{a_{n-1}}{n}t^n) = f$, and hence T is surjective. Thus the image of T is $\mathbb{P}_{n-1}$. We could also see this in the following way. Every element in $\mathbb{P}_n$ can be written as a linear combination of $1, t, t^2, \dots, t^n$, and hence every element in the image of T can be written as a linear combination of $T(1), T(t), T(t^2), \dots, T(t^n)$. Thus the image of T is the span of $T(1), T(t), T(t^2), \dots, T(t^n)$, which gives the image of T is the span of $1, 2t, 3t^2, \dots, nt^{n-1}$. Thus the image is all of $\mathbb{P}_{n-1}$.
- (c) Let $f \in \mathbb{P}_n$, then $f(t) = a_0 + a_1t + \cdots + a_nt^n$. Now $f \in \ker(T)$ if and only if $T(f) = 0$. Thus we get $T(f) = a_1 + 2a_2t + \cdots + na_nt^{n-1} = 0$. Equating coefficients, we get $a_1 = 0, 2a_2 = 0, \dots, na_n = 0$, and hence $a_1 = a_2 = \cdots = a_n = 0$. Thus a_0 is a free variable and all the other coefficients must be zero. Thus f is in the kernel if and only if $f(t) = a_0$. Therefore

$$\ker(T) = \{a_0 : a_0 \in \mathbb{R}\} = \text{Span}\{1\}.$$

6. (a) Again note that every element in $\mathbb{P}_2$ can be written as a linear combination of $1, t, t^2$, so the image of T will be given by linear combinations of $T(1), T(t), T(t^2)$. Note $T(1) = t$, $T(t) = t^2 + 1$, and $T(t^2) = t^3 + 2t$. Thus

$$\text{Range}(T) = \text{Span}\{t, t^2 + 1, t^3 + 2t\}.$$

- (b) Let $f \in \mathbb{P}_2$, then $f(t) = a_0 + a_1t + a_2t^2$. Now $f \in \ker(T)$ if and only if $T(f) = 0$. Now $T(f) = t(a_0 + a_1t + a_2t^2) + a_1 + 2a_2t = a_1 + t(a_0 + 2a_2) + a_1t^2 + a_2t^3$. Thus $T(f) = 0$ if and only if $a_1 = 0, a_0 + 2a_2 = 0, a_1 = 0$, and $a_2 = 0$. Thus all the coefficients of f must be zero, and hence $\ker(T) = \{0_{\mathbb{P}_2}\}$.

(c) **Claim:** T is one-to-one.

Let $f, g \in \mathbb{P}_2$, then $f(t) = a_0 + a_1t + a_2t^2$ and $g(t) = b_0 + b_1t + b_2t^2$. Suppose $T(f) = T(g)$, then $a_1 + t(a_0 + 2a_2) + a_1t^2 + a_2t^3 = b_1 + t(b_0 + 2b_2) + b_1t^2 + b_2t^3$. Equating coefficients, we get $a_1 = b_1$, $a_0 + 2a_2 = b_0 + 2b_2$, $a_1 = b_1$, and $a_2 = b_2$. Thus $f = g$, and hence T is one-to-one.

Claim: T is not onto.

We show that 1 is not in the image of T . Suppose 1 is in the image of T , then $1 = at + b(t^2 + 1) + c(t^3 + 2t) = b + (a + 2c)t + bt^2 + ct^3$ for some scalars a, b, c . Thus equating coefficients we get $1 = b$, $a + 2c = 0$, $b = 0$, and $c = 0$. But we have a contradiction since b cannot be zero and one! Thus 1 cannot be in the range of T and hence T is not surjective.

7. (a) Yes. T is linear, since

$$T(cp(t)) = \begin{bmatrix} cp(0) \\ cp(1) \end{bmatrix} = c \begin{bmatrix} p(0) \\ p(1) \end{bmatrix} = cT(p)$$

and

$$T(p + q) = \begin{bmatrix} (p + q)(0) \\ (p + q)(1) \end{bmatrix} = \begin{bmatrix} p(0) + q(0) \\ p(1) + q(1) \end{bmatrix} = \begin{bmatrix} p(0) \\ p(1) \end{bmatrix} + \begin{bmatrix} q(0) \\ q(1) \end{bmatrix} = T(p) + T(q).$$

(b) Let $f \in \mathbb{P}_2$, then $f(t) = a_0 + a_1t + a_2t^2$. Thus $T(f) = \begin{bmatrix} a_0 \\ a_0 + a_1 + a_2 \end{bmatrix}$. Thus

$$\text{Range}(T) = \left\{ \begin{bmatrix} a_0 \\ a_0 + a_1 + a_2 \end{bmatrix} : a_0, a_1, a_2 \in \mathbb{R} \right\} = \text{Span} \left\{ \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right\} = \mathbb{R}^2.$$

Now $f \in \ker(T)$ if and only if $T(f) = 0$. Thus $f \in \ker(T)$ if and only if $a_0 = 0$ and $a_0 + a_1 + a_2 = 0$. Thus $f(t) = -a_2t + a_2t^2$. Hence

$$\ker(T) = \{-a_2t + a_2t^2 : a_2 \in \mathbb{R}\} = \text{Span}\{t(1 - t)\}.$$

Thus the kernel of T consists of polynomials of degree 2 or less that have zeros at $t = 0$ and $t = 1$.

8. Note $T(1) = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$, $T(t) = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$, and $T(t^2) = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$. Thus

$$\text{Range}(T) = \text{Span} \left\{ \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \right\}.$$

Note this is different from $\text{Span} \left\{ \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \right\}$ since the entries along the diagonal can be different.

Now we compute the kernel of T . Let $f \in \mathbb{P}_2$, then $f(t) = a_0 + a_1t + a_2t^2$. Now $f \in \ker(T)$ if and only if $T(f) = 0_{M_n \times n}$, and hence we get $\begin{bmatrix} a_1 + a_2 & 0 \\ 0 & a_0 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$. Thus we get $a_0 = 0$ and $a_1 + a_2 = 0$. Thus $f(t) = -a_2t + a_2t^2$. Finally

$$\ker(T) = \{-a_2t + a_2t^2 : a_2 \in \mathbb{R}\} = \text{Span}\{t(1 - t)\}.$$